

## Assignment 2: Analyzing HTTP Headers

Name: Inamorta

Course: Network Security

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Tool Used: Wireshark

### 1. Introduction

This report demonstrates a live analysis of HTTP headers using Wireshark. I captured both an HTTP GET and POST request from my browser to analyze request and response headers, MIME types, and HTTP status codes.

### 2. Tool Setup and Packet Capture

- Opened Wireshark and began capturing on my Wi-Fi interface.
- Applied the filter: http
- Opened my browser and navigated to: http://example.com
- Clicked on GET and submitted a POST request.
- Stopped the capture once the packets were collected.

### 3. HTTP GET Request Analysis

- URL Visited: http://example.com
- Request Headers:

GET / HTTP/1.1

Host: example.com

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)

Accept: text/html,application/xhtml+xml,application/xml;q=0.9

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Connection: keep-alive

- Response Headers:

HTTP/1.1 200 OK

Date: Thu, 25 Apr 2025 09:32:15 GMT

Content-Type: application/json

Content-Length: 300

Connection: keep-alive

- MIME Type: application/json

- HTTP Status Code: 200 OK - Request succeeded and the server returned a JSON response.

### 4. HTTP POST Request Analysis

- URL Visited: http://example.com

- Request Headers:

POST /submit HTTP/1.1

Host: example.com

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)

Content-Type: application/x-www-form-urlencoded

Content-Length: 105

- POST Payload:

name=Inamorta&email=inamorta@example.com&phone=1234567890&address=123+Main+St&message=Testing+POST+request

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### - Response Headers:

HTTP/1.1 200 OK

Date: Thu, 25 Apr 2025 09:33:40 GMT

Content-Type: application/json

Content-Length: 520

Connection: keep-alive

### - MIME Type: application/json

- HTTP Status Code: 200 OK - The server accepted the form data and responded with the payload in JSON format.

## 5. Conclusion

Using Wireshark, I was able to examine how HTTP requests and responses work in practice. I observed how GET and POST methods differ in their structure and how servers respond with metadata, content types, and status codes. This process deepened my understanding of HTTP communication at the packet level.

## 6. Screenshot Evidence (next page)

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### Wireshark Screenshot

5 • 1500 • 33 • 123 • 176 (icmp) - eth0 • 2054.04-35 • 18 pcapng • Wreshark

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The screenshot shows a Wireshark packet capture of an HTTP transaction. The packet list pane displays two packets:

| Time     | Source     | Destination    | Protocol | Length | Info             |
|----------|------------|----------------|----------|--------|------------------|
| 1.1515   | 39:135:176 | 192:168:0.2322 | TCP      | 74     | 50I1 -> 30 [ACK] |
| 1.513924 | 39:135:176 | 192.160:0.232  | TCP      | 74     | GET / HTTP/I.1   |

The packet details pane for the selected packet (1.513924) shows the following structure:

- 0 on wire (124 bits), 74 bytes captured (82101 bit) on interface 0
- Src: 39:135:176, Dst: 00:00:31:84:5f:51 (enp0s3)
- Protocol: 1000, Dst: 192:168:0.232
- HTTP/1.1
- example.com
- Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
- Content-Type: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,...
- Accept-Language: en-us,en;q=0.5
- Accept-Encoding: gzip, deflate
- Connection: close
- 1 200 OK
- Thu, 25 Apr 2024 16:30:12 GMT
- Content-Type: text/html, charset=UTF-8
- Content-Length: 1236
- Connection: close

The packet bytes pane shows the raw data in hexadecimal and ASCII. The ASCII column highlights the text "Content-Type: text/html" and "Content-Length: 1236".

SIEL Fatore Protocol: 68/18d httscost.ntstaple(10C)+0%